

NYCMOS:

Selectable Homodyne/Sliding IF Downconversion Mixer

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Motivation

- Modern RF receivers use multiple downconversion schemes, each with different tradeoffs in complexity, frequency-planning, and low-frequency behavior.
- These concepts are often difficult to visualize in traditional RF coursework because intermediate signals are hidden inside RF systems.
- We propose an educational RF downconversion IC that exposes internal nodes and allows direct comparison of receiver architectures in real-time.

MOSBius Goals

- Design a configurable RF downconversion educational IC that allows students to directly compare receiver architectures.
- Demonstrate difference between homodyne and heterodyne mixers
- Sliding IF mixer can double as a normal heterodyne mixer as well, so students get exposure to 3 different downconversion approaches
- Expose intermediate outputs by passing them through buffers.
- Exposure to AM demodulation

Design

On Chip

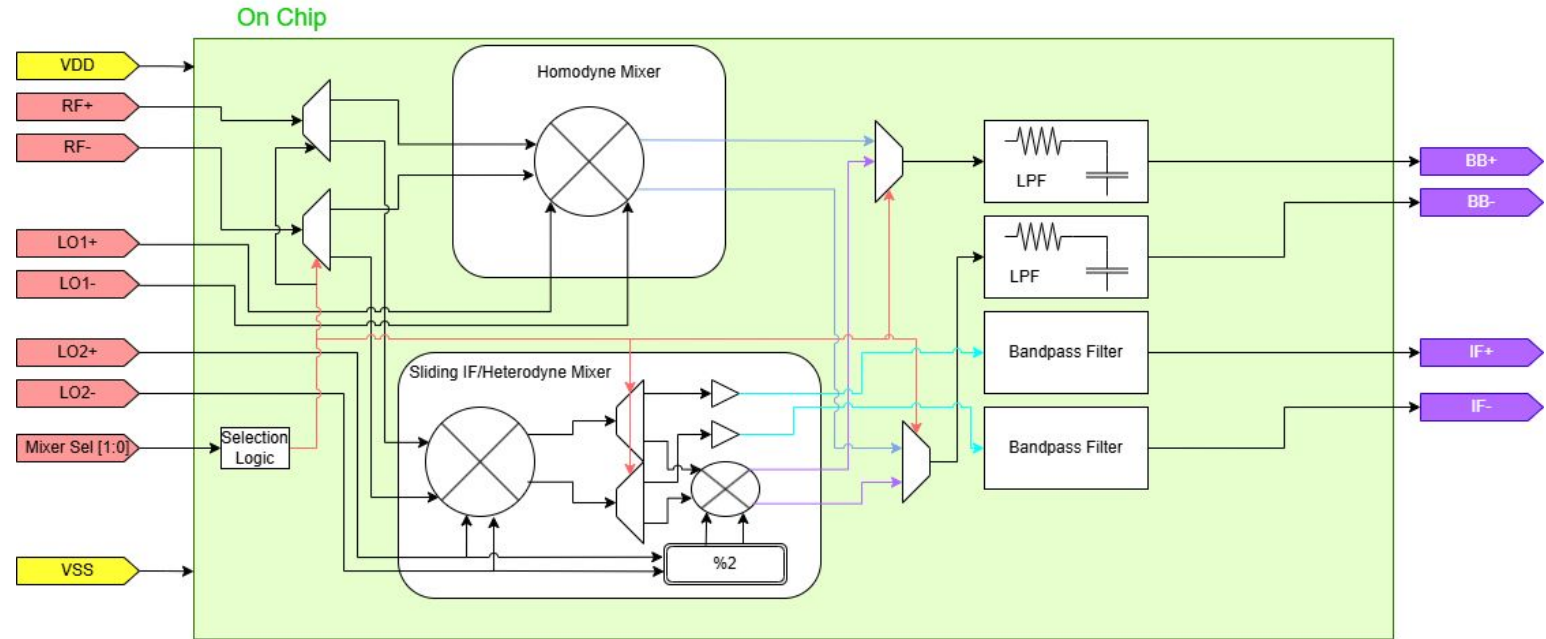
- Homodyne Downconversion Mixer
- Sliding IF Downconversion Mixer
 - Two mixers + 1 divider
- LPF for baseband output
- 4 mux switches to select which mixer is active, and which mixer output goes to the LPF
- 3 switches for picking between internal and external current mirrors for mixers
- 6 buffers to expose internal signals for measurement/observation

Design

Off Chip

- LO
 - Team Prof. Morbius from SSCS Chipathon 2026 used [Si535x-B20QFN-EVB](#)
- External Current Mirror
 - From Mosbius chip
- AM signal, generated via signal generator/external chip

Design



Measurements

- **Frequency Translations:** RF input mixes with LO to produce observable IF/baseband tones.
- **Conversion gain:** Increasing mixer current sink increases output amplitude.
- **Spectral content:** FFT shows desired tones, LO leakage, unwanted mixer products, with and without filtering stages.
- **LO Tuning:** Frequency mismatch shifts baseband and impacts recovered signal quality.
- **Architecture comparison:** Students compare different receivers by observing intermediate signals and demodulated outputs.

Team Background

We are graduate students in electrical engineering at NYU with a concentration in analog/RF circuit design.

Demian - Experience: RF front-end design in Cadence Virtuoso.

Lei - Experience: RF/Analog design in cadence virtuoso. Amplifiers/LO design.

Charbel - Experience: Electrical Engineer at Amogy, RF front-end design and SpectreRF simulation, Research Assistant at the Laboratory for Laser Energetics.

References

- Razavi, B. (2011). *RF Microelectronics* (2nd ed.). Prentice Hall.